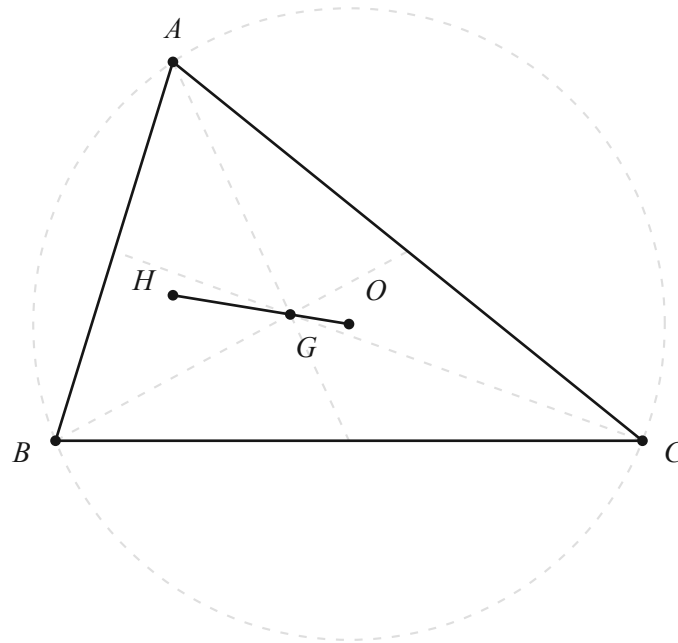


A field guide to triangle centres

Three familiar constructions lead to three special points. Their connection is one of the most satisfying first results in triangle geometry.



Centroid G. The medians join each vertex to the midpoint of the opposite side. Their common point divides each median in the ratio 2:1, measured from the vertex.

Circumcentre O. The perpendicular bisectors meet at the centre of the circle through all three vertices.

Orthocentre H. The altitudes are the lines through the vertices perpendicular to their opposite sides. Their common point is H.

The Euler line, in vectors

Assume ABC is non-degenerate. Place the origin at its circumcentre O, and let a, b, c be the position vectors of A, B, C. All have the same length R .

$$|a|^2 = |b|^2 = |c|^2 = R^2$$

The centroid has vector $g = (a + b + c)/3$. Indeed, the point two-thirds of the way from A to the midpoint of BC has exactly this vector; the same holds for the other medians.

A candidate for the orthocentre

Set $h = a + b + c$. The vector from A to this point is $b + c$. The direction of BC is $b - c$. Their dot product is

$$(b + c) \cdot (b - c) = |b|^2 - |c|^2 = 0.$$

Thus the candidate lies on the altitude from A. Cycling the letters proves that it lies on all three altitudes. It is H.

Read off the conclusion

We have $h = 3g$. Since O is the origin, O, G, H are collinear and G is one-third of the way from O to H. For a non-equilateral triangle:

$$OG : GH = 1 : 2.$$

For an equilateral triangle all three centres coincide. The vector identity is still true, but these points no longer determine a unique line.

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An example, and two exercises

Worked example. Take $A = (0,3)$, $B = (-2,0)$, $C = (2,0)$. Symmetry puts O on the vertical axis. Write $O = (0,t)$ and equate squared distances to A and B :

$$(3 - t)^2 = 4 + t^2, \text{ so } t = 5/6.$$

Hence $O = (0,5/6)$, $G = (0,1)$, and $H = 3G - 2O = (0,4/3)$. The distances $OG = 1/6$ and $GH = 1/3$ give the promised ratio. In arbitrary coordinates, use $H = A + B + C - 2O$.

Exercises

1. In a right triangle, identify H and O without calculation. Describe the Euler line.
2. For $A = (0,0)$, $B = (6,0)$, $C = (0,8)$, calculate O , G , and H . Verify the vector identity.

Solutions

1. H is the right-angle vertex, because the legs themselves are altitudes. O is the midpoint of the hypotenuse. The Euler line joins those two points.

2. $O = (3,4)$, $G = (2,8/3)$, $H = (0,0)$. Then $3G - 2O = (6,8) - (6,8) = (0,0)$. The lengths are $OG = 5/3$ and $GH = 10/3$.

Further reading in the journal: "Three centres. One remarkable line." These notes give an original exposition of a classical result.